

Arnav Sharma

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EDUCATION

University of California, Davis

Expected Jun 2027

B.S. in Computer Science · Minor in Business Studies

Dean's List: Spring 2024

Relevant Coursework: Machine Learning, Artificial Intelligence, Computer Vision, Operating Systems, Computer Architecture

EXPERIENCE

Fonabit AI

Jun 2026 – Sep 2026

Software Engineer Intern

- Developed custom Model Context Protocol (MCP) servers and APIs integrating LLMs with enterprise CRM databases.
- Designed multi-agent orchestration using Agent-to-Agent (A2A) protocols for autonomous sales and lead-management workflows.
- Built AI security gating, prompt-injection defense, and RBAC to block unauthorized data access, plus human-in-the-loop guardrails requiring manual approval for high-stakes agent actions.
- Optimized prompt chaining and context-window retrieval, lowering token overhead and cutting response latency.

Pixabits Technologies Pvt. Ltd.

Jun 2025 – Aug 2025

Software Developer Intern

- Built and tuned backend services and REST APIs in Python and Java, cutting response time and raising throughput by over 50%.
- Profiled and resolved performance bottlenecks with cross-functional teams, shipping fixes across multiple releases.
- Translated Figma designs into responsive UI components, iterating on user analytics to lift engagement.

PROJECTS

Verity — Agentic Credit-Note & Reconciliation Platform

2026 · Site · Demo

- Built and deployed an agentic accounting platform that ingests a company's bank feed, classifies each payment, issues the credit owed, and posts it back — 19 containerized services live, ~870 tests passing.
- Made the LLM advisory, never authoritative: proposals must clear schema, invoice, and policy validation, and an AST-level test proves no agent can import an issuance capability.
- Enforced multi-tenant isolation in PostgreSQL (FORCE row-level security, NOBYPASSRLS), with tests asserting unscoped queries fail closed rather than leak across tenants.
- Guaranteed money correctness with idempotency keys, a transactional outbox, and claim-with-lease drainers, so concurrent workers and crashes cannot double-post.
- Shipped Zoho Books (OAuth 2.0), Plaid, and ISO 20022 camt.053 behind one adapter contract, plus a React/TypeScript operator UI and an Astro marketing site.

CardioSense

2026 · HackDavis · GitHub

- Built a sub-\$30 wearable ECG that continuously detects atrial fibrillation on-device (no cloud), returning a rhythm verdict in under 10 seconds.
- Trained a 43K-parameter 1D CNN (PyTorch) on MIT-BIH clinical ECG data, reaching ~91% accuracy (95% with a strict "Uncertain" gate) at ~10 ms inference.
- Built the signal-processing pipeline (filtering, Pan-Tompkins R-peak detection, HRV features) and real-time SwiftUI/Streamlit interfaces over BLE.

InfraCopilot AI

2026 · UC Davis · GitHub

- Built a full-stack predictive-maintenance platform for EV charging networks with a cost-aware ML model (scikit-learn, SMOTE) achieving ~90% failure recall and \$300K+ simulated savings.
- Engineered a FastAPI backend and Next.js dashboard delivering real-time risk scoring and root-cause insights.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Haskell, Erlang, Prolog, Assembly

Machine Learning & AI: PyTorch, scikit-learn, CNNs, Transformers, LLM Agents (Claude/Gemini API, LangGraph, MCP)

Paradigms & Systems: Functional Programming, Logic Programming, Concurrency, Operating Systems, Memory Management

Backend & Tools: PostgreSQL (RLS), FastAPI, Docker, REST APIs, React/TypeScript, Git, Linux, pytest